

Lab 10: Exam Academic year 2022-2023

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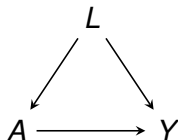
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Question 1 - Setup

- Average causal effect of A on Y .
- We have covariates L sufficient to adjust for confounding (and possibly more).
- Use *random forest regression* to predict Y based on L , separately for $A = 1$ and $A = 0$ (one model for the treated, one for the controls).
- Average the difference of the predictions

$$\frac{1}{n} \sum_{i=1}^n \hat{E}_i^{rf}[Y \mid A = 1, L] - \hat{E}_i^{rf}[Y \mid A = 0, L]$$

Question 1 (a) - Appropriate Selection of Confounders



- When fitting the two outcome models, the selected confounders are those that help predicting the outcome Y .
- However, there may be some confounders that are **strongly** associated with the exposure A and only a small association with the outcome Y .
- In this last case, those confounders **may be excluded** from the outcome model, thus leading to an incorrect selection of confounders.

Question 1 (a) - Appropriate Selection of Confounders

		Y		
		Zero	Small	Strong
A	Zero	Irrelevant	Irrelevant	Irrelevant
	Small	Irrelevant	Greyzone	Important
	Strong	Irrelevant	Important	Essential

Table: Appropriate selection of confounders based on the relationship between L and A , Y . (Table taken from Michael Knaus).

- Relying only on the outcome model, we may miss those confounders having a **small** association with Y but **strong** with A .

Question 1 (b) - Standard Error with Bootstrap

- When using **data adaptive** estimators, there is a **bias-variance tradeoff** to minimize the MSE.

$$\text{MSE} = \text{Bias}^2 + \text{Variance}$$

- To minimize the MSE we want to balance the two terms, i.e., they should converge at the same rate.
- For most data-adaptive estimators, the optimal trade-off leads to a bias $O(n^{-\alpha})$, with $\alpha < \frac{1}{2}$.

Mean Squared Error

The **mean squared error** of an estimator $\hat{\theta}$ for θ is

$$\text{mse}[\hat{\theta}] = \mathbb{E} \left[(\hat{\theta} - \theta)^2 \right].$$

By expanding the square we find that

$$\begin{aligned} \text{mse}[\hat{\theta}] &= \mathbb{E} \left[(\hat{\theta} - \theta)^2 \right] \\ &= \mathbb{E} \left[(\hat{\theta} - \mathbb{E}[\hat{\theta}] + \mathbb{E}[\hat{\theta}] - \theta)^2 \right] \\ &= \mathbb{E} \left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2 \right] + 2\mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]](\mathbb{E}[\hat{\theta}] - \theta) + (\mathbb{E}[\hat{\theta}] - \theta)^2 \\ &= \text{var}[\hat{\theta}] + (\text{bias}[\hat{\theta}])^2. \end{aligned}$$

Question 1 (b) - Standard Error with Bootstrap

- But when we shift focus from the nuisance function to the target parameter (e.g., the ATE), we find:

$$\text{Bias} = O(n^{-\alpha}), \quad \text{Standard Error} = O(n^{-1/2})$$

- Since $\alpha < 1/2$, the bias **dominates** the standard error, so even in large samples, confidence intervals will not be centered around the true parameter.
- Even using the bootstrap, confidence intervals constructed using standard error estimates will be **biased**, because the bootstrap does not correct for systematic bias — only variability.
- This motivates the use of **debiased estimators** such as TMLE or AIPW to enable valid inference.

Question 1 (c) - Debiased Estimators

Since the bias affecting a plug-in estimator of a model-free estimand is primarily driven by minus the sample average of the **estimand's efficient influence curve**...

- Derive the efficient influence function of the target parameter (i.e., the ATE).
- Use a debiased estimator that sets its sample average to zero.
- Valid inference can be obtained by $(1/\sqrt{N})^* \text{sd}(\text{EIC})$.

Question 2 (a) - Setup

We consider Assumption-Lean **Logistic** Regression:

$$\text{logit} \{E(Y^a | L)\} = \alpha(L) + \beta a$$

- Y : dichotomous outcome.
- A : dichotomous exposure (randomly assigned).
- L : baseline characteristics.

Question 2 (a) - Assumption-Learn Logistic Regression

For a dichotomous exposure A , we consider weighted averages

$$\frac{E\{w(L)\beta(L)\}}{E\{w(L)\}}$$

with $\beta(L) = g\{E(Y | A = 1, L)\} - g\{E(Y | A = 0, L)\}$.

We know that the treatment is **randomized**, thus $w(L)$ is constant.

The estimand reduces to

$$\frac{E\{w(L)\beta(L)\}}{E\{w(L)\}} = E[\beta(L)]$$

Question 2 (a) - Assumption-Lean Logistic Regression

Note that $\beta(L)$ is defined as:

$$\begin{aligned}\beta(L) &= \text{logit}(E[Y^1 \mid L]) - \text{logit}(E[Y^0 \mid L]) \\ &= \log(\text{odds}(Y = 1 \mid A = 1, L)) - \log(\text{odds}(Y = 1 \mid A = 0, L)) \\ &= \log\left(\frac{\text{odds}(Y = 1 \mid A = 1, L)}{\text{odds}(Y = 1 \mid A = 0, L)}\right)\end{aligned}$$

Thus, in the randomized setting (constant weights), the assumption-lean logistic regression targets the average conditional log-odds ratio, i.e., $E[\beta(L)]$.

- If $\beta(L)$ is **constant**, homogeneous treatment effect on the log odds scale.
- If $\beta(L)$ is **not constant**, using assumption-lean logistic regression we are still targeting the mean of the conditional log-odds ratio.

Question 2 (b) - Non Collapsibility

Is the average of the conditional log-odds ratio equal to the marginal log-odds ratio?

$$E \left[\log \left(\frac{\text{odds}(Y = 1 \mid A = 1, L)}{\text{odds}(Y = 1 \mid A = 0, L)} \right) \right] = \log \left(\frac{\text{odds}(Y = 1 \mid A = 1)}{\text{odds}(Y = 1 \mid A = 0)} \right)$$

- **Non-collapsibility** is a property of some statistical measures of association that causes them to change when adjusting for a covariate, even if that covariate is not a confounder.
- Other measures are **collapsible**: for instance, risk differences.

Question 2 (b) - Non Collapsibility

- Assume that $\beta(L) = \beta$, i.e. the treatment effect is constant. This means that **all the conditional log-odds ratio are the same**.
- However, the **marginal log-odds ratio can still be different** due to non-collapsibility: conditioning on covariates changes the magnitude.
- This happens due to the 'mathematical' way the (log-)odds ratio are defined.

Question 2 (b) - Non Collapsibility

Conditional and marginal odds ratios

	Counterfactual risk (placebo)	Counterfactual risk (treatment)	Causal odds ratio
Men (25%)	0.5	0.75	3.0
Women (75%)	0.25	0.5	3.0
Overall	0.3125	0.5625	2.83

Table: Huitfeldt A, Stensrud MJ, Suzuki E. On the collapsibility of measures of effect in the counterfactual causal framework. *Emerg Themes Epidemiol.* 2019 Jan 7;16:1. doi: 10.1186/s12982-018-0083-9. PMID: 30627207; PMCID: PMC6322247.

Noncollapsibility of the OR is the phenomenon whereby the crude OR from the marginal table cannot be expressed as the weighted average of the stratum-specific ORs even in the absence of confounding.

Question 2 (b) - Non Collapsibility

If instead we look at risk differences:

$$\beta(L) = E(Y \mid A = 1, L) - E(Y \mid A = 0, L)$$

$$\begin{aligned} E[\beta(L)] &= E[E(Y \mid A = 1, L) - E(Y \mid A = 0, L)] \\ &= \underbrace{E[Y \mid A = 1] - E[Y \mid A = 0]}_{\text{marginal RD}} \end{aligned}$$

- Unlike the odds ratio, the **risk difference is collapsible**.
- Averaging the conditional risk differences over the population recovers the marginal risk difference.

Question 2 (c) - Estimating the propensity score

- $0.5 \rightarrow$ Yes, since we know it by design.
- $\sum_{i=1}^n A_i/n \rightarrow$ Yes, because we know it is constant (in an observational study we could not estimate the propensity score with a sample average).
- $\hat{E}^{rf}(A | L) \rightarrow$ Yes, since in the general case we would estimate it using data-adaptive estimators.

Question 3 (a) - Orthogonal Learners

Disadvantages of T-Learner

- It **does not target** the CATE, but instead consists of two separate minimization problems resulting in two, possibly regularized, outcome predictions.
- Risk of **extrapolation** if the distribution of the covariates is different for treated and controls.
- It is **not orthogonal** (see next slide to understand).

T-Learner

The prediction problems do not know of joint goal to approximate a difference

$$\Rightarrow \hat{\mu}_1(X) \text{ minimizes } \text{MSE}(\hat{\mu}_1(X)) = \mathbb{E} \left[(\hat{\mu}_1(X) - \mu_1(X))^2 \right]$$

$$\Rightarrow \hat{\mu}_0(X) \text{ minimizes } \text{MSE}(\hat{\mu}_0(X)) = \mathbb{E} \left[(\hat{\mu}_0(X) - \mu_0(X))^2 \right]$$

BUT what they should aim to minimize is

$$\begin{aligned} \text{MSE}(\hat{\tau}(X)) &= \mathbb{E} \left[(\hat{\tau}(X) - \tau(X))^2 \right] \\ &= \mathbb{E} \left[(\hat{\mu}_1(X) - \hat{\mu}_0(X) - (\mu_1(X) - \mu_0(X)))^2 \right] \\ &= \mathbb{E} \left[(\hat{\mu}_1(X) - \mu_1(X))^2 \right] + \mathbb{E} \left[(\hat{\mu}_0(X) - \mu_0(X))^2 \right] \\ &\quad - 2\mathbb{E}[(\hat{\mu}_1(X) - \mu_1(X))(\hat{\mu}_0(X) - \mu_0(X))] \end{aligned}$$

Question 3 (a) - Orthogonal Learners

Advantages of Orthogonal Learners (DR-Learner, R-Learner,...)

- They **directly target the CATE** by minimizing the counterfactual prediction error:

$$E \left[\left\{ Y^1 - Y^0 - \psi(L) \right\}^2 \right]$$

- They are **orthogonal**, meaning that even if you use data-adaptive estimator for nuisance parameters, you have asymptotically the same properties as if they were given.
- DR-Learner is orthogonal because it relies on the **efficient influence function of the ATE**.
- R-Learner achieves orthogonality by **residualization**.

Question 3 (b) - Orthogonal Learners in Randomized Experiments

In both the orthogonal learners, there are 2 nuisance parameters:

- Outcome model
- Propensity score

If we are in a randomized setting, there is **no need to estimate the propensity score**, since it is constant and known.

This is especially useful in the context of DR-Learner, since it relies on the positivity assumption of the propensity score.

Question 3 (c) - (Naive) MSE among the Treated

We are given $\psi(L)$ for $E(Y^1 | L)$. We would like to estimate

$$E \left[\{Y^1 - \psi(L)\}^2 \right]$$

as the sample average of $\{Y - \psi(L)\}^2$ among treated individuals.

Is it valid?

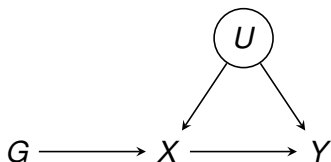
- In an **observational study**, the characteristics of the treated and controls are different. Thus estimating this quantity only on the treated subsample would not give a valid estimate of the marginal mean squared error.
- However, in a **randomized setting**, it would be valid (note that in this case also the $ATE = ATT$).

Question 3 (d) - (Debiased) MSE

- Let $Z^1 = (Y^1 - \psi(L))^2$
- We are interested in estimating $E[Z^1]$
- The efficient influence function is

$$\frac{A}{P(A=1|L)} \{Z - E[Z | A=1, L]\} + E[Z | A=1, L] - E[Z^1]$$

Question 4 (a) - Causal Diagram



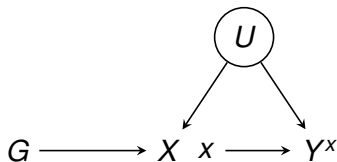
where

- Y = Alzheimer's disease (outcome),
- X = Diabetes (exposure),
- G = HLA-DR3 genotype (IV),
- U = ... (unobserved confounders)

Question 4 (b) - Test in presence of unmeasured confounders

- We want to test whether diabetes affects the risk of Alzheimer's disease.
- However, since we have unmeasured confounders, we can not carry out a naive analysis between those two variables.
- We know that the effect of the IV is *only through* the exposure.
- We can run a simple linear regression of $Y \sim G$ and test whether its coefficient is different from 0.

Question 4 (c) - SWIG and Independence Assumption



The independence assumption:

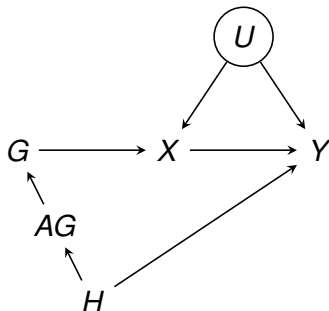
$$Y^x \perp\!\!\!\perp G$$

says that there is no association between the outcome and the IV in the absence of the exposure.

Question 4 (c) - SWIG and Independence Assumption

- Is that assumption testable?
- No, since Y^x is a counterfactual, thus independence is not empirically testable. It is an assumption required for identification using G as an IV.

Question 4 (d) - Covariates H



where

- Y = Alzheimer's disease (outcome),
- X = Diabetes (exposure),
- G = HLA-DR3 genotype (IV),
- U = ... (unobserved confounders)
- H = History of Alzheimer's disease (confounders)
- AG = Ancestral Genotypes (mediator of the effect of H to G)

Question 4 (d) - Covariates H

- H is a confounder of G and Y , thus not controlling for it **violates** the independence assumption.
- We will revisit the assumption as **conditional** independence:

$$Y^x \perp\!\!\!\perp G \mid H$$

Question 4 (d) - IV Estimation

- Consider the model

$$E(Y | X, H, G, U) = \psi X + g(H, U)$$

- We can define ψ by the **estimand**:

$$\psi = \frac{E\{\text{Cov}(G, Y | H)\}}{E\{\text{Cov}(G, X | H)\}}$$

If instead the model is **misspecified** (ψ is not constant):

$$E(Y | X, H, G, U) = \psi(H)X + g(H, U)$$

our estimand reduces to

$$\psi = \frac{E\{\text{Cov}(G, Y | H)\}}{E\{\text{Cov}(G, X | H)\}} = \frac{E\{w(H)\psi(H)\}}{E\{w(H)\}}$$

for weights $w(H) = \text{Cov}(G, X | H)$.

Question 4 (d) - IV Estimation

We can then estimate ψ using debiased machine learning:

- **Efficient influence curve**

$$\frac{\{G - E(G | H)\}[Y - E(Y | H) - \psi\{X - E(X | H)\}]}{E[\{G - E(G | H)\}\{X - E(X | H)\}]}$$

- **A debiased estimator that sets its sample average to zero**

$$\hat{\psi} = \frac{\sum_{i=1}^n \{G_i - \hat{E}(G_i | H_i)\} \{Y_i - \hat{E}(Y_i | H_i)\}}{\sum_{i=1}^n \{G_i - \hat{E}(G_i | H_i)\} \{X_i - \hat{E}(X_i | H_i)\}}$$

- **Valid inference** can be obtained by the usual $\frac{1}{\sqrt{N}} * \text{sd}(\text{EIC})$

Question 4 (d) - IV Test

A better way to test whether diabetes affects the risk of Alzheimer's disease is obtained by testing:

$$\theta = E\{\text{Cov}(G, Y \mid H)\}$$

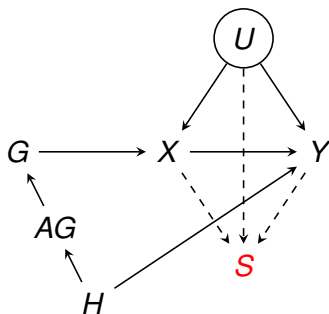
The efficient influence curve of θ is

$$\{G - E(G \mid H)\}\{Y - E(Y \mid H)\} - \theta.$$

Which leads to

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \left\{ G_i - \hat{E}(G_i \mid H_i) \right\} \left\{ Y_i - \hat{E}(Y_i \mid H_i) \right\}$$

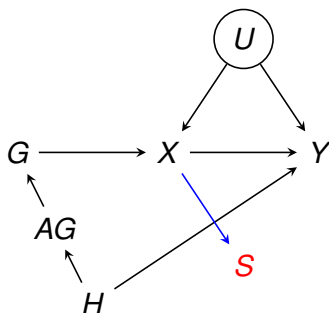
Question 4 (e) - Selection Bias



where

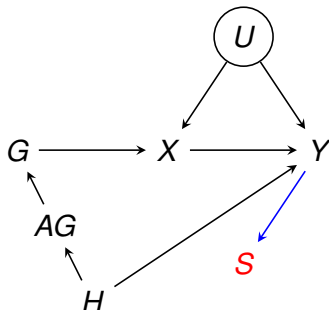
- S = Binary Live Status
- Other variables as before

Question 4 (e) - Selection Bias (i)



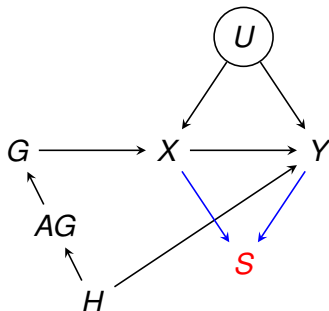
- S is a child of X (which is a collider of G and U).
- Conditioning on S would be similar to conditioning on X , thus creating a spurious association between G and U .
- This would cause bias since there will be a path from G to Y through the unobserved factors U , leading to a violation of the exclusion restriction (regardless of whether X affects Y).

Question 4 (e) - Selection Bias (ii)



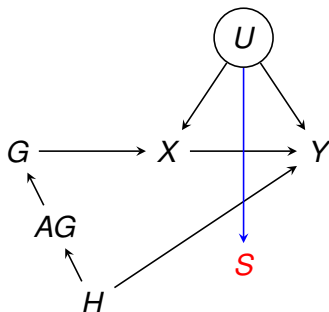
- If $X \longrightarrow Y$, then S is a **collider** that would cause bias in the estimation of the effect. Note that S is affected by G (and also by X) through Y , and is affected clearly by Y .
- If $X \nrightarrow Y$, then S is not a collider anymore since it will be not affected by G and X .

Question 4 (e) - Selection Bias (iii)



- Conditioning on S induce **collider bias**.
- If there was no relationship between X and Y , conditioning on S may induce it.
- If there is an effect from X to Y , conditioning on S may introduce bias in the estimation.

Question 4 (e) - Selection Bias (iv)



- Since S is not affected by other variables, conditioning on it is not problematic for the analysis.